

STEPGEN · CALCULATION NOTE

StepGen sizing calculations

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DOCUMENT CONTROL

Document ID	SGN-CAL-001	Version	0.3
Status	Draft	Date	2026-10-01
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stepgen		
Source	docs/04-calcs/01-sizing.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-01	Amish Chadha	Re-run for the constructable design (SGN-DDR-003); mass, geometry, lean clearance, cost and cradle standoff updated; cost reported against the value-engineering target
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). Deck rails 60 x 30 x 2 mm; R12 checked on the salvage reference build; key switch fitted; all results re-run
0.1	2026-09-25	Amish Chadha	First TRL 3 sizing note (geometry, mass, road load and range, hill and acceleration, belt, control timing, braking and structure, SwapCell interface v0.3, cost)

StepGen sizing calculations

On paper the walking-treadmill vehicle meets its range, speed, braking, step-off, power and SwapCell interface requirements, and its cost is under the value-engineering target. Version 0.2 applied the recommendations Amish accepted on 2026-09-25 (SGN-DDR-002): 60 x 30 x 2 mm deck rails and the salvage reference build. Version 0.3 re-runs every result for the constructable design of SGN-DDR-003: the frame, guards and fixings change the mass slightly (35.1 kg, 37.9 kg with the pack), the vehicle is 2.38 m long with its fender and 0.58 m wide, and the reference build costs USD 610 against a USD 650 value-engineering target (USD 750 with all new parts). Three requirements are **at risk**: R1 (a 1.00 m belt fits a 1.75 m rider only up to 5 km/h; the belt stays at 1.05 m until co-design), R6 (the 8 % hill needs 235 W, a 6 % margin to the 250 W rating) and R10 (37.9 kg with the pack, a 2.1 kg margin that is smaller than a 10 % mass-growth allowance). Guarding (R9) and pack retention to SwapCell latch class V1 (R13) cannot be verified until hardware exists, which is TRL 4 work and on hold by Amish's instruction.

Every number in this note is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`), which also writes `docs/04-calcs/results.csv`. The script takes its geometry from `cad/src/model.py`, its prices from `bom/bom.csv` and its budget from `project.yaml`, so the note, the model and the BOM share one source. All values are first-principles estimates; nothing here is measured.

■ 1. Assumptions

Table 1. Inputs. All are assumptions for a paper design.

Input	Value	Basis
Design rider	80 kg, 1.75 m, centre of mass 0.56 of height above the belt	Typical adult
Heaviest rider (structure)	120 kg	Upper design mass
SwapCell pack	2.85 kg; 457 Wh at the terminals per full cycle; 10 % reserve kept	SWC-CAL-001 (interface v0.3)
Rolling resistance	Crr 0.010	20 x 1.75 in tires on pavement
Air drag	CdA 0.70 m ² , air density 1.2 kg/m ³	Standing rider, larger than a seated upright cyclist
Drivetrain	Motor and gear 80 %, controller 95 %	Geared hub at part load
Auxiliary load	3 W	Logic board, display, controller standby
Pack voltage	46.8 V nominal, 39.0 V minimum	SwapCell 13S
Real-use energy factor	x 1.4	Stops, hills, wind; indicative only
Belt over the roller bed	Rolling coefficient 0.02 on the rider's weight	Belt indentation over 30 mm rollers
Belt bending and bearings	5 N bending at two end rollers; bearing friction coefficient 0.0015	Estimates
Belt tension	500 N per run (tensioner setting)	Chosen in section 7
Step and foot length	Step 0.40 x height at 5 km/h and 0.45 x height at 6 km/h; foot 0.15 x height	Typical adult gait; to be checked in co-design
Tire to road friction	0.7	Dry pavement
Steel	S235 class, yield 235 MPa, E 200 GPa; welded detail FAT 71 with partial factor 1.35	Eurocode 3 style check
Walking load	1.2 x body weight per step; 1.9 steps/s, 1 h a day for 5 years	Estimate
Class V1 environment	8 g sine, 25 g shock, 3.5 kg design pack	SwapCell interface v0.3 item V

■ 2. Geometry and mass

The model gives a vehicle 2379 mm long including the rear fender (2349 mm over the tires), 580 mm wide at the bar, with a 1855 mm wheelbase. The belt runs on end rollers 1050 mm apart, leaving 1000 mm of usable length, 400 mm wide, with its top 240 mm above the ground. Ground clearance under the cross members is 145 mm. The bar is 1230 mm above the ground (990 mm above the belt). A 70° head angle with a 30 mm fork offset gives 58.0 mm of trail and 18.6 mm of wheel flop, in the range of ordinary bicycles. The TRL 2 model had no fork offset, which gave about 90 mm of trail.

Table 2. Mass estimate.

Item	Mass (kg)	Basis
1 Main frame	12.7	Deck rails 6.21 (60 x 30 x 2 mm), two cross members 1.44, nose beam 1.51, four rear stays 0.92, down tube 1.16, machined head tube 0.52, dropouts, caps, lugs, tabs and plates 0.90
2 Belt and end rollers	3.8	Belt 1.3, two rollers 2.2, axles and tension bolts 0.3
3 Roller bed	2.9	14 rollers at 0.15 kg, two aluminium carrier bars 0.8
4 One-way bearing and drag screw	0.1	The bearing replaces a roller bearing
5 Belt speed sensor	0.05	Estimate
6 Rear wheel with hub motor	4.3	Motor 2.5, rim, spokes, tire and tube 1.6, rotor 0.2
7 Front wheel, fork, headset	3.0	Wheel 1.6, fork 1.1, headset 0.3
8 Steering column, bar, grips	1.7	38 x 2 mm column, stem, bar, grips
9 Brakes	1.1	Two calipers, rotors, levers, cables
10 Receiver cradle, class V1	1.0	3 mm steel cradle, lever, receptacle
12 Controller and logic board	0.6	Estimate
13 Display, selector, lanyard, key	0.3	Estimate
14 Guards	2.1	Heel, toe and roller guards in aluminium sheet, HDPE side boards, fender
15 Harness	0.4	Estimate
16 Hardware and kickstand	1.0	Kickstand 0.4, fasteners and rivet nuts 0.6
Vehicle without pack	35.1	
SwapCell pack	2.85	SWC-CAL-001
Vehicle with pack	37.9	R10 limit 40 kg

The margin to 40 kg is 2.1 kg, or 6 % of the vehicle mass; a 10 % growth allowance would take the vehicle to 41.4 kg, so R10 is **at risk** on mass. The larger deck rails (SGN-DDR-002) add 1.08 kg. With an 80 kg rider the total is 117.9 kg. The vehicle's centre of mass is 909 mm ahead of the rear axle and 330 mm up; the rider's is at 950 mm and 1220 mm; together 937 mm ahead and 934 mm up, with 49 % of the static load on the rear wheel.

■ 3. Road load, energy and range

Wheel power is rolling plus air drag, $P = (C_{rr} m g + \frac{1}{2} \rho C_d A v^2) v$. Pack power is wheel power over 0.76 plus 3 W. Usable energy is 411 Wh.

Table 3. Steady speed on the flat, 117.9 kg.

Speed	Rolling	Air	Wheel	Pack	Current at 46.8 V (39.0 V)	Energy	Range, flat	Real use
15 km/h	48 W	30 W	79 W	106 W	2.3 A (2.7 A)	7.1 Wh/km	58 km	about 41 km
20 km/h	64 W	72 W	136 W	182 W	3.9 A (4.7 A)	9.1 Wh/km	45 km	about 32 km
25 km/h	80 W	141 W	221 W	294 W	6.3 A (7.5 A)	11.8 Wh/km	35 km	about 25 km

At 20 km/h the controller loses about 9 W and the motor and gear about 34 W. Road load equals the 250 W rating at 26.4 km/h, so the 25 km/h assist cut-off, not the motor, sets top speed. At the 250 W wheel cap the pack supplies 332 W: 7.1 A at 46.8 V, 8.5 A at 39.0 V, well inside the 15 A controller limit. **R5 is met:** 45 km against 30 km at 20 km/h.

■ 4. Hill, acceleration and motor

An 8 % grade at 8 km/h needs 106 N at the wheel, which is 235 W (a 6 % margin to 250 W), 26.1 N m at the hub and 6.7 A from the pack. At 250 W the vehicle climbs a 5 % grade at 12.1 km/h, 8 % at 8.5 km/h and 10 % at 7.0 km/h. R6 is met, but the margin is thin, so it is marked **at risk**.

The wheel turns at 268 rpm at 25 km/h and 86 rpm at 8 km/h, so the hub motor must be wound for a 20 in wheel: about 400 rpm no-load at 48 V, which leaves about 325 rpm at 39 V. To hold acceleration from rest to 1.0 m/s² (R3), the firmware limits drive force to 129 N, a hub torque of 32.0 N m. With the 250 W cap the vehicle reaches 20 km/h in about 12 s and 25 km/h in about 27 s; the available acceleration falls from 1.00 m/s² at 5 km/h to 0.64 at 10 km/h, 0.17 at 20 km/h and 0.04 at 25 km/h. The vehicle cannot surge away from a walking rider.

■ 5. Belt drag, walking power and belt length

The push needed to walk the belt at 5 km/h is 15.7 N for the roller bed, 5.0 N for belt bending and 1.0 N for the end roller bearings, plus 0 to 8 N from the drag setting: **21.7 to 29.7 N**, inside the 30 N of R1 even at full drag. Walking puts 30 to 41 W into the belt at 5 km/h, and 24 to 49 W over 4 to 6 km/h, all of it dissipated as drag. A slider deck with a friction coefficient of 0.1 to 0.2 would need 78 to 157 N, which confirms the roller bed.

The belt length is the weak point of R1. With the step and foot assumptions of Table 1:

Table 4. Belt length needed (step plus foot) against 1.00 m usable.

Rider	5 km/h	6 km/h
1.60 m	0.88 m, fits	0.96 m, fits
1.75 m	0.96 m, fits	1.05 m, does not fit
1.90 m	1.04 m, does not fit	1.14 m, does not fit

R1's target of 1.0 m is met exactly, but the target itself is short for tall riders or brisk walking, so R1 is **at risk**. A 1.90 m rider at 6 km/h needs a roller pitch of about 1190 mm, which would make the vehicle about 2.52 m long with its fender and break R10 (2.4 m). Decided by Amish, 2026-09-25 (SGN-DDR-002): keep the 1.05 m roller pitch and 2.35 m length over the tires (2.38 m with the fender of SGN-DDR-003), and revisit after co-design shows the riders' heights and pace.

For comparison, if 40 W of walking drove the wheel with no loss at all, the vehicle would reach only 9.8 km/h on the flat, and a 3 % grade at 5 km/h needs 65 W. This is why the motor does the work. The optional roller generator (not fitted, SGN-DDR-001 item 5) would return 5 to 10 W, a charge of 0.11 to 0.21 A in SwapCell mode 4, and about 5 % more range at 20 km/h.

■ 6. Control timing (R2)

The 8-magnet ring on the 50 mm front roller gives a pulse every 19.6 mm of belt, or every 47 ms at the 1.5 km/h start threshold, so the 0.5 s start window sees 11 pulses. When the belt stops, the logic board declares a stop after three missed periods (141 ms), then adds 10 ms of logic, a 100 ms controller ramp-down and 10 ms of current decay: **261 ms from belt stop to motor off**, inside the 500 ms target. The brake-lever switches and the lanyard switch are wired to the controller's brake-cut input, so they act in about 30 ms. At 25 km/h the vehicle coasts 1.8 m in 261 ms. R2 is **met on paper**; the logic exists only as a labeled sketch.

■ 7. Braking, stability and structure

At 3 m/s² the vehicle stops from 25 km/h in 8.0 m after the brakes act (R7, 10 m). With the combined centre of mass 937 mm ahead of the rear axle and 934 mm up, the rear brake alone can give 2.51 m/s² (9.6 m) and the front alone 5.36 m/s² (4.5 m), so either brake alone still stops within 10 m. The vehicle would pitch over only at 9.6 m/s² if the rider were fixed to the deck; the practical limit is the rider, who must resist 240 N at 3 m/s².

The one-way bearing inside the rear roller (SGN-DDR-003) holds it, so the rider can brace on the belt. With 180° of wrap and a friction coefficient of 0.3 the capstan ratio is 2.57, so the belt needs 273 N of tension per run not to slip under 240 N; the tension bolts are set to 500 N. The bearing sees only 6.0 N m against about 56 N m rated for its CSK17 class.

The lowest outboard part when leaning is now the kickstand body on the left rail, which gives 32.6° of lean clearance: a minimum turn radius of 2.8 m at 15 km/h and 7.7 m at 25 km/h. (The concept's outboard clutch housing, which set 30° before, is gone.)

The deck rails are now 60 x 30 x 2 mm RHS on edge ($I = 159,499 \text{ mm}^4$, $Z = 5317 \text{ mm}^3$), decided by Amish on 2026-09-25 (SGN-DDR-002). They carry a 120 kg rider at 2.5 g at mid span at 80 MPa, a factor of 3.0 on yield, with 1.5 mm of deflection. Each step gives a 25.5 MPa stress range, about 12.5 million times in five years of daily use, against 38.8 MPa for a FAT 71 weld with a partial factor of 1.35, so the weld details now have a margin. The previous 50 x 25 x 2 mm rails (version 0.1) gave 37.6 MPa, at the limit; the change adds 1.08 kg, which reduces the R10 mass margin. There is no numbered requirement for structure; this is reported for the review.

The steering column must carry the rider's push at the bar, taken as 500 N at 449 mm above the head tube. A 32 x 2 mm column reaches 169 MPa (factor 1.4); the chosen **38 x 2 mm column** reaches 116 MPa (factor 2.0).

■ 8. SwapCell interface v0.3

StepGen uses the interface unchanged (R11, R13).

- **Wake (item W).** The receiver fits a $10\text{ k}\Omega \pm 1\%$ coding resistor between INTERLOCK and SGND. With the pack's $100\text{ k}\Omega$ pull-up to 3.3 V the node sits at 0.297 to 0.303 V for 9.90 to $10.10\text{ k}\Omega$ (plus $0.2\text{ }\Omega$ of wiring and switch), inside the 0.24 to 0.37 V window, and the loop draws $30\text{ }\mu\text{A}$. A key switch sits in series with the resistor (decided by Amish, 2026-09-25, SGN-DDR-002; for use at a standstill only): key off opens the loop, the pack opens its output within 1 ms and sleeps after 60 s at $100\text{ }\mu\text{A}$ (about 0.72% per month); key on pulls the node below 1.0 V and wakes the pack with no supply from the vehicle. The WAKE pin is not used.
- **Heartbeat.** The logic board is the SwapCell host: host type 0 (vehicle), requested mode 2 (discharge), host discharge limit 15.0 A . In use the pack supplies 3.9 A at a 20 km/h cruise and at most 8.5 A , inside the 15 A legacy limit, and the pack makes only about 8.0 W of heat at that maximum ($110\text{ m}\Omega$), far from the 20 A case that puts SwapCell R3 at risk. Every circuit stays below 60 V DC .
- **Charge-discharge mode (item C).** Not used. The geared hub freewheels, so there is no regenerative braking, and the roller generator is not fitted.
- **Latch class V1 (item V).** For a 3.5 kg design pack, 8 g gives 275 N and 25 g gives 858 N . The pack latch proof is 1717 N (1.72 kN). The receiver must preload the pack against its end stop with 330 N or more, which an over-centre lever with a ratio of 6.6 gives at a 50 N hand force. The cradle's two M6 8.8 bolts see 21 MPa in shear at 25 g (a factor of 22 on 480 MPa), and the $44 \times 2\text{ mm}$ down tube sees 24 MPa from the shock at the pack's 75 mm standoff. Retention under vibration can only be shown by test, so R13 is **not verifiable at TRL 3**.
- **Flag to SwapCell.** The vehicle's logic board is powered from the pack output, so after a key-on wake the pack will enter legacy discharge (no heartbeat within 2 s) before the board can send one. The v0.3 behavior rules do not say whether a pack in legacy discharge moves to mode 2 when a heartbeat arrives. This is flagged to the SwapCell project, not changed here.

■ 9. Cost

All 16 BOM lines are priced at new-part prices. Amish decided on 2026-09-25 (SGN-DDR-002) to make the salvage route the reference build, with new parts as the fallback, and on 2026-10-01 that `budget_usd` is a hypothetical value-engineering target, not a limit. Value-engineering target: USD 650. Estimated cost of the constructable design: **USD 610 on the reference build** (USD 40 under the target), pack excluded. The reference build uses a used walking-pad treadmill for items 2 and 3 at about USD 40 and a donor 20 in bike for the front wheel and brakes at about USD 50; the donor's 20 in fork is too short for the head tube (SGN-DDR-003), so the fork and headset are bought, about USD 40. The all-new-parts fallback is USD 750, USD 100 over the target; it is reported, not held to R12. Making the design constructable added USD 64 to the reference build (USD 546 before).

■ 10. Results against requirements

Table 5. Requirement status. "At risk" means met on paper with less than 10 % margin or resting on an unverified assumption.

ID	Value	Target	Status
R1	Usable belt 1.00 m x 0.40 m; push 22 to 30 N at 5 km/h; 1.00 m fits a 1.75 m rider only up to 5 km/h	1.0 m, 0.36 m, 30 N at 5 km/h	At risk
R2	Belt stop to motor off 261 ms; brake and lanyard about 30 ms; start after 11 pulses in 0.5 s	Start after 0.5 s above 1.5 km/h; off within 0.5 s	Met (on paper)
R3	Cut at 25 km/h, beginner 15 km/h; torque limit 32 N m gives 1.0 m/s ²	25 km/h, 15 km/h mode, 1.0 m/s ² or less	Met (on paper)
R4	250 W rated motor specified; wheel power capped at 250 W	250 W rated or less	Met (on paper)
R5	45 km at 20 km/h on the flat (9.1 Wh/km); about 32 km in real use	30 km at 20 km/h	Met
R6	235 W at the wheel for 8 % at 8 km/h (6 % margin)	8 % at 8 km/h within 250 W	At risk
R7	8.0 m at 3 m/s ² ; rear brake alone 9.6 m; belt held by the one-way bearing with 500 N tension	Two brakes; 10 m from 25 km/h; belt cannot run forward	Met (on paper)
R8	Belt top 240 mm; side boards 9 mm proud of the belt, no rails	250 mm or less; open sides	Met
R9	Guards modelled at both nips, the rear tire and belt ends; gaps not yet checked against ISO 13857	Nips, spokes and rear tire guarded	Not verifiable at TRL 3
R10	2.38 m long with the fender, 0.58 m wide, 37.9 kg with the pack (2.1 kg margin)	2.4 m, 0.65 m, 40 kg	At risk
R11	Interface v0.3: 10 kΩ coded INTERLOCK (node 0.30 V), vehicle heartbeat mode 2; 8.5 A maximum; below 60 V	SwapCell v0.3 unchanged; 15 A or less; below 60 V DC	Met (on paper)
R12	Reference (salvage) build USD 610 excluding the pack, USD 40 under the value-engineering target; all-new-parts fallback USD 750	USD 650 value-engineering target excluding the pack on the reference build	Under the target (on paper)
R13	Class V1 receiver specified: preload 330 N, lever ratio 6.6, bolt factor 22 at 25 g	SwapCell latch class V1, no release or contact break	Not verifiable at TRL 3

■ 11. Checks against the documents

The TRL 2 figures were checked against this note and the documents were corrected where they differed: vehicle mass 35 kg became 34.0 kg (36.8 kg with the heavier 2.85 kg pack), 9.0 Wh/km became 9.1 Wh/km once the 3 W auxiliary load was included, range 46 km became 45 km, pack power 179 W became 182 W, walking power 20 to 40 W became 30 to 41 W at 5 km/h, the 8 % hill 236 W became 233 W, the no-loss mechanical drive 7 to 9 km/h became 9.9 km/h, trail went from about 90 mm (no fork offset) to 58 mm, and the parts cost \$630 became \$713. Version 0.2 (SGN-DDR-002) moved the vehicle to 35.0 kg (37.9 kg with the pack), the hill to 235 W and the cost to \$546 on the reference build (\$721 all new). Version 0.3 (SGN-DDR-003) moves it to 35.1 kg, 2.38 m long, 0.58 m wide, 145 mm ground clearance, 32.6° lean clearance and USD 610 on the reference build (USD 750 all new).

Safety: These are paper sizings for a vehicle a person stands and walks on, driven by a 468 Wh lithium-ion pack at up to 54.6 V DC. The braking, belt-slip, fatigue, steering-column and latch figures above are not a substitute for testing, which is TRL 4 work and on hold. Nothing here clears the design for building or riding.